Jayasurya Marasani

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Education

Politecnico di Milano – M.Sc. in Computer Science and Engineering **2025 – Present** | *Milan, Italy*
Amrita Vishwa Vidyapeetham – B.Tech. in CS (Artificial Intelligence) | CGPA 9.09/10 **2020 – 2024** | *Kerala, India*

Technical Skills

Languages: Python, Java, SQL, JavaScript, MATLAB
ML / DL: Deep Learning, Computer Vision, NLP, Speech Recognition, Multimodal Learning, Vision-Language Models, Transformers (ViT), LLMs (GPT, BERT), Embedding Models & Retrieval Evaluation (MRR, Recall@K), Generative AI, Deep RL (DQN, PPO, A2C), TinyML, Quantization, Ensembles, XGBoost, Random Forest, Semantic Segmentation, Pose Estimation
Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Keras, TensorFlow Lite Micro, ONNX, scikit-learn, OpenCV, Stable-Baselines3, Gym, NumPy, pandas, SciPy
Cloud / DevOps: Azure (OpenAI, Functions, App Service, Blob Storage, Storage Queue, Cosmos DB), GCP, CI/CD (GitHub Actions), Containerization, Linux
Backend / Data: FastAPI, Django, React, REST APIs, Microservices, SQL, MongoDB, Cosmos DB

Experience

Project Manager & ML Engineer, Polimi Data Scientists (PMDS) **Feb 2026 – Present**
Politecnico di Milano student data science association *Milan, Italy*
- **Leading the Maxillary Canine Impaction project**, designing an **anatomy-prompted ROI attention network** for automated impaction prediction from panoramic dental radiographs
- Building a **multi-task deep learning pipeline** (localization, classification, clinical risk scoring) with **landmark-guided feature fusion** for interpretable 2D dental imaging
- Developing an **on-device sleep-phase detection model** for the BEA Smart Sleep Band, engineered for **256 KB RAM** and a $<$**50 KB model size** for real-time local inference

Junior Developer – AI & Immersive Technologies **Oct 2025 – Feb 2026**
Ophir Studio *Milan, Italy*
- Prototyped **generative AI solutions across text, image, video and 3D**, and delivered low-code **MVPs** for internally developed products
- Developed **APIs and microservices** for scalable digital platforms and implemented **DevOps workflows** (CI/CD pipelines, containerization, cloud deployment)
- Partnered with creative teams and universities to **integrate AI into storytelling products**

Undergraduate Research Assistant & Team Lead **Jan 2022 – Aug 2024**
Amrita Vishwa Vidyapeetham *Kerala, India*
- **Led a team of 5** from problem framing to publication, producing **9 peer-reviewed papers** (Elsevier, IEEE) across computer vision, NLP, speech and deep reinforcement learning
- **Initiated an international collaboration** with Konkuk University (Seoul), leading 3 researchers to build DRL agents for image classification that reached **97.57% on MNIST** and **outperformed ML/DL baselines**
- Built a **multi-agent DRL ensemble** (DQN, PPO, Recurrent PPO, A2C) with weighted voting, reaching **98.55% accuracy on Galaxy Zoo**
- Developed a **multimodal pose model** fusing MoveNet and ResNet50 via attention; introduced the **YFA-30 dataset** and reached **96% accuracy**
- Built a **VGG16 + XGBoost** gaze classifier (**98.48% accuracy**); benchmarked **LLMs (PaLM, GPT, BERT)** for topic classification and **ASR models (Whisper, Wav2Vec2, SpeechBrain)**

Projects

Depthwise Spatio-Temporal Network for UWB Radar Localization | *TinyML, TFLite Micro, ESP32-S3 Apr 2026 – Aug 2026*
- Built a physics-guided **TinyML** model (**22K parameters**, depthwise separable residual blocks) for device-free localization of up to 4 people from raw CIR data of 6 UWB radars
- Reached **F1 0.943** (recall 0.982) with **0.21 m mean error** using the fully **INT8-quantized** model, shrunk to **42.2 KB** with 209 KB activation memory to fit ESP32-S3 limits

EmbedMatch: Embedding Model Selection for Retrieval | *Azure, FastAPI, React, ONNX* *Mar 2026 – Apr 2026*
- Built a benchmarking platform that ranks **embedding models on a user's own dataset**, comparing **Azure OpenAI** with local **ONNX** models (MiniLM, BGE) on **MRR, Recall@K**, relevance and latency via a weighted composite score
- Designed an **asynchronous serverless pipeline** using Azure Blob Storage, Storage Queue, Azure Functions and Cosmos DB

Selected Publications *(9 peer-reviewed, 1 under review; full list at jayasurya-marasani.github.io)*

- Harnessing Deep Reinforcement Learning Algorithms for Image Categorization, *Engineering Applications of AI* (Elsevier, IF 8.0), 2024
- Multi-Agent Deep RL for Autonomous Galaxy Image Classification, *Egyptian J. Remote Sensing & Space Sciences* (Elsevier), 2026
- Multi-Modal Hybrid Yoga Pose Recognition with Attention and Late Fusion on YFA-30, *International Journal of Yoga*, 2025
- An Empirical Analysis of Topic Categorization Using PaLM, GPT and BERT Models, *IEEE i-PACT*, 2024
- Speech Recognition Paradigms: Comparing SpeechBrain, Whisper and Wav2Vec2 Models, *IEEE I2CT*, 2024